UG Stage 2 Team Project (IN2033) module 2025/2026
All BSc and MSci students in the second year of their studies, excluding students on Professional Pathway

Developing a Software Product

The InfoFarma Ordering System (IPOS)

Student's Brief

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Version 1.3 released on 18th of February 2026
- Replaced Martin Walter as Consultant of 9 teams (Team 7 - 15) with Dr Kaleem Peeroo

Version 1.2 released on the 10th of February 2026
- Updated the Team membership by highlighting two Professional Pathway students, not taking IN2033, who were initially included in Teams.
- Added a footnote resolving the apparent discrepancy between the summary of report IPOS-SA-RPT iii) and the layout provided in Appendix 5.

Version: 1.1 released on the 6th of February 2026.
Minor typos fixed in this version of the document.

Version: 1.0 *was released on the 28th of December 2025.*

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1. Introduction

1.1 Purpose and Scope

This document is the Student's Brief about the Team Project module (IN2033). It describes the overall organisation of the module and the activities that each team are supposed to undertake. It presents project timetable, defines the module assessment, and details what each team and each student must deliver at each stage of the project. The document contains team membership, too, together with the initial statement of requirements about the case-study that will be used throughout the project.

All students on the BSc (all routes) and MSci (all routes) degree programmes in the Computer Science department undertake the team project in their second year of the UG studies. The only exception to this are the students on the Professional Pathway route (they attend IN2034 Work-Based Project instead). Team Project module involves elements of analysis, design, implementation and testing of a software system to meet a given set of requirements.

A ***purpose*** of the project is to give the students an appreciation of the challenges and difficulties of working with peers in a team to satisfy the requirements of a ***fictitious commercial customer*** using the software engineering methods taught on the BSc/MSi Programmes to solve a realistically complex problem (that is, one that cannot be solved by an individual student in the time available). More specifically, the students are expected to use the knowledge and skills gained in Stage 1 (e.g. IN1007 Programming in Java and IN1013 Databases modules) as well as the in Stage 2 module IN2013 Object-Oriented Analysis and Design.
Another equally important purpose set out in the module is enhancement of ***transferable skills*** including teamwork, negotiation and interpersonal skills. The chosen case study – an IT system of significant complexity (InfoPharma Ordering System, IPOS) – offers scope for collaboration not only with teams, but also between teams. The IPOS system is split into ***three sub-systems*** which address distinctly different business needs and are expected to interact as a complete system with data exchanged between the sub-systems. Each sub-system will be developed by a separate team, three teams will form a group, and the teams in the same group are expected to work in collaboration to deliver a fully functional IPOS prototype. This collaborative arrangement will provide an opportunity for each student to not only learn to take responsibility within a small software development team, but also to collaborate with other development teams to formulate the software architecture (high-level design) of the entire system and to spell out in detail how the sub-systems will interact with one another ***before*** the subsystems are developed[1]. This arrangement mimics the industrial practice as software architecture is typically decided at the start of software development and ***remains unchanged*** throughout the stages of development (analysis, design, implementation and testing of the subsystems).

The module runs in Term 2. Week numbers used in the module (and in this document) are counted from the first teaching week of Term 2 and extend for 12 consecutive weeks until the end of the term (including the Reading/Project week in the middle). Some submission deadlines are scheduled for week 12. The rationale is to allow the students to maximise their success.

Although this document outlines the tasks that each team must perform, it does not mandate any particular method of planning – this is left for the students to choose and follow. Within the module a lecture will cover the basics of ***agile methods*** of software development, e.g. Scrum, Kanban, etc., with examples of how to apply ***elements of agile development*** in the project. For instance, students will be encouraged to consider learning about ***user stories*** as a way of capturing system's functional requirements, which will allow them to appreciate the difference from alternative methods of dealing with functional requirements such as Volere Template and Use case models.

For part of the development, the teams/groups are encouraged to use UML and other methods formally taught during their courses (Java programming and database design). The teams are expected to exhibit a degree of initiative and to seek and apply solutions they ***have not already been taught***. Being prepared to learn new things independently and experiment with them is essential for professional development throughout one's professional life.

All groups of teams will undertake the project based on the *same case-study* but must work independently!

[1] Extensive literature exists on software architecture, which you are encouraged to look at. This is often referred to as high-level design with many non-agile software development methods. Using agile methods does not remove the need for software architecture. On the contrary, many see software architecture, which must be decided early, as one of the few decisions which should remain unchanged throughout the development despite the flexibility of agile methods which embrace rapid pace and changes that typically take place throughout agile software development.

1.2 Learning Objectives

The learning objectives are to demonstrate the ability to plan, organise and conduct a software development project of realistic complexity, to document the software development (e.g. with UML), and deliver a software package within schedule, and of adequate quality to meet customer requirements.
The ability to collaborate (i.e. to work together and effectively), with the other members of the same team and with other teams in the same group, to plan realistically, and to meet deadlines are important factors affecting success.
Specifically, apart from enhancing the relevant technical skills, the students will gain experience of the following:

- forming a working team, establishing common goals and resolving misunderstandings and conflicts.
- communicating effectively with other stakeholders such as the customer and the other teams in the group, responsible to deliver parts of a complex software product.
- planning time and effort and allocating tasks to individuals.
- collaborating with other teams on formulating high-level design (software architecture) of a complex software product consisting of several sub-systems, developed in parallel by different development teams.
- identifying and resolving risks and problems.
- establishing and using document filing and configuration standards, version control software.
- working to a life-cycle with defined deliverables[2].

1.3 Prerequisites

Students are expected to draw upon material taught and experience gained in the Stage 1, as well as Stage 2 courses taught in parallel with Team Project. The following modules are particularly relevant:

- IN1007 – Programming in Java (Stage 1)
- IN1013 – Databases (Stage 1)
- IN2013 – Object Oriented Analysis and Design (Stage 2). This is a **co-requisite** module.

1.4 Recommended reading

For most of the UML tasks especially relevant reading are as follows:

- Jim Arlow, Ila Neustadt, UML 2.0 and the Unified Process: Practical Object-Oriented Analysis and Design, Addison-Wesley, 2nd edition, July 2005, 592 pages, ISBN 0321321278
- M. Fowler, UML Distilled: a brief guide to the standard object modelling language, 3rd Edition, Addison-Wesley, 2003
- G. Booch et al, Object-Oriented Analysis and Design with Applications, 3rd Ed., Addison-Wesley, 2007
- S. Bennett et al, Object-Oriented Systems Analysis and Design Using UML, 4th Ed., McGraw Hill, 2010
- B. McLaughlin et al, Head First Object-Oriented Analysis and Design: A Brain Friendly Approach to OOAD, O'Reilly, 2006
- Cay, S. Hortsmann, *Object-Oriented Design & Patterns*. Hoboken, NJ, Wiley. 2006
    - This text is useful especially to learn about design patterns, but also as a practical guide that links UML models with programming in Java.

For supplementing, and advancing, your Java knowledge and skills the following is useful:

- C. Horstmann, *Big Java*, 7th edition, "Relational Databases" Chapter, John Wiley and Sons.
    - http://www.horstmann.com/bigjava.html

For furthering your knowledge and skills in Human Computer Interaction, the following is recommended:

- D. Benyon, *Designing Interactive Systems,* 4 Edition, Pearson, 2019

Relevant web resources are as follows:

- UML Specification
    - http://www.omg.org/spec/UML/2.5.1/
- Object Management Group, UML Resource Page
    - http://www.uml.org/
- Practical UML: A Hands-On Introduction for Developers
    - http://edn.embarcadero.com/article/31863
- Allen Holub's Quick UML Reference
    - http://www.holub.com/goodies/uml/
- Java tutorial on JDBC
    - http://download.oracle.com/javase/tutorial/jdbc/

[2] Although the project may include elements of agile software development and dedicated teaching sessions on agile methods will be provided in the module, the teams are ***not expected*** to adopt a specific agile method (e.g. Kanban, Scrum, etc.). The development process model may include elements of agile development (e.g. to document functional requirements using user stories), but it will include two project deliverables, which is more aligned with managed software development processes (e.g., Waterfall, Unified Process and many others).

Please see also the reading materials suggested in the related modules above. Further information about the relevant literature for the module can be found at: http://readinglists.city.ac.uk/index.html . See also the Library Guide for the CS department - Computing Lib Guides.

2. Organisation

2.1 Teams

Before the official start of the module, the students are assigned to teams of 7 (+/- 1) members, all of whom will be on one of the department's BSc/MSci degree courses. The size of the teams was decided based on many years of successful running of the module – the team size is chosen to allow a **degree of redundancy** so that in case a team member withdraws, or lacks commitment, this does not necessarily lead to the team failure. The rationale behind the pre-assignment of students to teams is to ensure that teams have a **balance of talent, skills and expertise** and become aware that in their future careers one must learn to establish good working relations with "strangers" as well as acquaintances and friends (see sections "1.1 Purpose and Scope", and "1.2 Learning Objectives" for further explanations).
Allocation to the teams is done **based on the marks achieved** in Stage 1 (on IN1007), and only *secondarily* based on coursework marks in IN2013. The students ranked in the top, middle and bottom of the cohort are **equally** represented in each team. This ensures as fair as possible team membership for *all* students.
The students may **not** change the team once assigned (no expulsions, no poaching etc.).

A ***team consultant***[3], who is a member of the Team Project Management Team (TPMT), is allocated to each team. The consultant role(s) are described below – see section "2.4 The Consultant".

Each team will work independently of others on a part of the same case-study. The internal organisation of each team, planning the project, defining a management structure, carrying out the work (technical or organisational) etc., is the *joint responsibility of all the members* of the team, and nobody else. Each team member is expected to adopt ***two roles***, with associated job titles, within the team and to take responsibility for those tasks falling within their remit, from the following list:

- ***Project manager***, responsible for team planning, co-ordination, and risk management; he/she acts as primary interface to the team consultant and the customer (NB this person **must** contribute technical work too!).
- ***Deputy project manager***, second in command, and responsible for documentation, reports and standards (NB: this person must contribute technical work too!).
- ***System analyst***, responsible for elicitation of customer requirements and requirements analysis and specification.
- ***Designer***, responsible for the system design process and design specification.
- ***Programmer***, responsible for implementing the system.
- ***Tester***, responsible for writing a test plan, testing the system and its components and recording the outcomes.

Each team member will have one ***primary role*** and a ***secondary role***, e.g., project manager/systems analyst or designer/programmer. The roles must be appropriately spread. This will ensure that essential roles within the team are covered when members are absent. There can only be one project manager role. A student cannot be both project manager and deputy, of course. An example of a typical configuration of roles within a team might be:

- Sarah: project manager/programmer
- Bob: deputy project manager/systems analyst
- Jim: designer/tester
- Ashraf: designer/programmer
- Tamara: systems analyst/programmer
- Nilesh: designer/tester
- Aisha: designer/programmer.

Despite the roles assigned to each member, it is strongly suggested that **all students be actively involved in, and contribute to, the work *throughout the project*** (though not necessarily with same level of involvement and contribution in each phase).

[3] In the specialised literature on agile software development methods, you may come across the term "Agile Coach", with broad responsibilities which ranging from coaching development teams to training the entire organisation, including senior management, to adopt agile methods including methods of effective communication, collaboration within the teams/organisation. Before practicing as Agile Coaches, the individuals are expected to undertake training course and secure accreditation/certification. The Team Project Management Team considered adopting the role of "Agile Coach" in the project but eventually decided against it. The main reason for this decision is to avoid giving the students the impression that what they are going to see from members of staff helping the teams in the projects (mainly to resolve difficulties in communication and lack of commitment, which often arise) is what professional Agile Coaches do and can offer.

Teams are encouraged to follow a software development *process* with elements of agile methods, structured around ***two project deliverables***. Also, the teams **must** use ***UML***, to document the software architecture (high-level design) and the detailed software design. For software requirements the teams are free to choose between using UML use case models and user stories used in agile software development.
The teams are free to choose their programming language, database platform, operating system and machine type (provided that *they have the right kit, e.g. laptops etc., themselves* in order to demonstrate the software to the TPMT during the "Demo" day). Traditionally, most teams implement their software using Java, for reasons of familiarity, and support availability[4]. However, teams are free to use other programming languages, e.g., C++, Python, C# or any of the other languages supported by the .NET platform, etc.

Teams must operate autonomously. ***All*** students should be present at ***all briefing, teaching and Q&A sessions***. Members of staff will be available to answer questions at every Q&A session.

Each team must keep a file of minutes of all their meetings, and a record of all tasks completed, documents (or parts of documents) delivered including the "rough", draft versions, etc. (as either hard-copies or e-copies). This content should be stored in the ***Project Binder,*** which is also referred to as ***Project Manual***. Minutes from meetings with the other teams in the same group, e.g. drafts of software architecture and of the interfaces of communication between sub-systems must also be taken and kept in the Project Binder/Manual.

In addition, ***each student must*** keep an ***Individual Diary*** of their own involvement in the project. An ***individual plan must be drawn up*** at the beginning of the project showing the ***expected hours*** to be put in during the project, broken down by tasks. Then, a record should be kept of the ***actual effort*** (in a time unit of choice, usually hours) spent and a reason should be given for any substantial discrepancy. In addition to helping students contribute to the project in an effective way, this will be an input to the Individual Report, which is a piece of the formal, summative assessment due in the end of the project (see below).

Both the Project Binder and the Individual Diaries must be shown to the respective team consultant ***on demand****. In any case intermediate versions must be submitted to the respective consultant at the end of each software development phase!* The team consultant will assess the individual contribution of each member of the team ***at the end of the project***, based upon these documents as well as other evidence e.g., attendance and involvement at the meetings with the consultant, the Individual Reports (see section "6.4Individual Final Report (Due: Week 12, 25% of total project marks)below), etc.

2.2 Groups

The need to create an environment of team collaboration, which mimics working practices in software industry, led to the selection of a case study of significant complexity – the IPOS (see section 8. InfoPharma Ordering System) – which is too complex for a single team to complete within Term 2.

Groups of ***three teams*** are, therefore, setup to work in collaboration to develop a fully functional prototype of the IPOS system. The teams in the group are labelled Team A, Team B and Team C, and will focus on developing one of the subsystems of IPOS. The three teams will collaborate throughout the project on the following aspects:

- define together the IPOS system software architecture including the necessary interfaces between the subsystems already defined in section 8. The outcome of this collaborative work will be a precise ***definition of the interfaces*** needed for communication between the IPOS subsystems to meet the customer requirements. The concept of "provided" and "required" interfaces, as used in UML component diagrams, will be used. These definitions will be further elaborated by developing a fragment of UML design class diagrams in which the interfaces and their respective implementation classes will be defined for each subsystem. ***The teams in the group must have consistent models of IPOS software architecture.***

- each team will conduct a ***peer – review*** of the first deliverable "Requirements Specification and High-Level System Design", produced by ***another team in the same group***. Peer reviews are expected to be objective and constructive, and to acknowledge the strength and weakness of the work completed by the other team.

- At the end of the project, the teams will exchange with each other ***pieces of software code***, which will be used in the prototype of the teams ***to access the functionality provided by the other subsystems***.

The membership of teams in groups is assigned as follows: Group 1 consists of Teams 1 (Team A), 2 (Team B) and

[4] Using Java as the programming language for the development brings additional benefits, since some parts of assessment require reverse engineering of the source code of the completed product into a design class diagram. Visual Paradigm Professional Edition, available at City, supports reverse engineering with Java and C++, but not with other programme languages.

3 (Team C), Group 2 consists of Teams 4 (Team A), 5 (Team B) and 6 (Team C), …, Group 16 consists of Team 46 (Team A), 47 (Team B) and 48 (Team C). Team/Group membership is provided in section 10. Teams, Groups and consultants. This membership is non-negotiable.

2.3 The "Customer"

Your project is to deliver a software package to meet the requirements of a "customer". Staff will play the role of the Customer (Info Pharma Ltd) representatives: Mr. Lancaster (founder and CEO), Mr Walter (CTO) and Mr Khan (Head of IT).
The teams in the same group will have to discover the full requirements by interviewing the Info Pharma representatives. The initial statement of requirements is included in this document (see section "8. InfoPharma Ordering System").
The interviews are set for Groups and are expected to be attended by representative of all three teams in the Group. Attendance by the *entire teams* is welcome but is not required. You are expected to have read carefully and analysed the text in section 8, and the material in section 9 before going for interviews with the representatives of the customer! Attention should be paid to both the functionalities of the subsystem your team is responsible for, and to the interaction between the subsystems.

The customer will evaluate some of the documents the teams deliver, including the final product. The other documents will be evaluated by the team consultant (or other members of the TPMT) and via peer review by another team in the same Group.

2.4 The Consultant

The **primary role** of the team consultant is to monitor the progress of the team and help resolve difficulties and risks communicated to him or her.
Please, note that not all consultants are equally versed in every part of the Team Project tasks. Team consultants **are not meant to act as technical consultants** to the teams. This is, partly, so that TPMT remain fair to all teams. If any technical advice is given by a consultant, it is the team's decision to use it or not. Technical questions, instead, are meant to be directed to the Customer/lecturer(s).

2.5 Communication

Most communication during the Team Project will be done by e-mail or Moodle. Therefore, the students must ensure that they are able to send and receive e-mail and check their e-mail inboxes, and Moodle messages, regularly.
There will be at least 4 categories of messages exchanged during the project:

1. *Internal messages between members of the team*. How to organise internal communication is up to the team members. Some may be deemed strictly private. Others will need to be filed in the Project Binder to record progress or setbacks.
2. *Messages between the members of a Group*. These messages are expected to be related to discussion of matters related to collaboration between the teams in the same group.
3. *Messages between the team and the customer*. These may have "contractual" significance, and teams should use a procedure for filing them for future reference. Most of them will need to be included in the Project Binder.
4. *Messages between the team and the Consultant*. These should be dealt with separately from customer messages. (They will normally be kept confidential from the customer.).
5. *Messages from the Team Project module leader* will deal with general administrative matters. For this the News Forum or the Discussion Forum accessible by all students will be used.

The teams are also encouraged to use Moodle for internal team communication (see http://moodle4.city.ac.uk/). Please note that each team is provided with a ***private Discussion*** Forum on Moodle. The team members, the respective consultant, and the lecturer(s) have access to a given Discussion forum. Private Discussion forums will be set up for each group, too. These forums will be accessible by all students in the group and their consultants.

Meetings are also a vital means of communication, and essential for success. There are, at least, the following meeting types: i) internal team meetings, ii) team consultant meetings, and iii) inter-team meetings, i.e. of meetings between representatives of the teams in the same group, to discuss matters related to IPOS architecture and sharing software for access the functionality across subsystems. Teams should hold the internal meetings regularly at appropriate frequency and arrange to see their consultant. Meetings with consultant are expected to be *weekly* in person[5]. The meetings between teams are expected to be called when needed (e.g. at the start of project to discuss architecture and towards the end of the project to help each team with the access to the functionality of the subsystem developed by the other teams). Please note that team consultants are very busy (they are all lecturers) and will need sufficient notice of appointments. A consultant may only be able to grant appointments at certain times/days but will see each of the teams for ***up to* 30 minutes every week**. For consultant meetings, the most effective way is to have

[5] This is in part a Visa compliance requirement.

a dedicated day of the week, and time, agreed in advance between the team and the consultant. The consultant meetings are **mandatory**! Consultants may be invited by the teams to attend the inter – team meetings, but their presence at these meetings is not mandatory.

3. Project Timetable

The project will extend throughout the whole of the teaching Term 2 (26 January – 18 April 2026), including the Reading, i.e. Project, week. Some project deliverables (i.e. documents, which will be formally assessed) will be due in **week 12, after teaching finishes**, to give you sufficient time to succeed. The deadlines have been agreed and optimised with the lecturers of all other Stage 2, Term 2 modules, in order to give you appropriate time to succeed in all modules. Moreover, all coursework deadlines have been published in the beginning of the academic year. The timetable of activities and deliverables (outputs) is given in Table 1 below. The contact hours and face-to-face assessments are on the official timetable too.

Week	Briefings/Q&As/Deadlines	Activities	Deliverables
1	(week 1) **26 January:** *Teaching Session:* • Briefing on module organisation • Briefing on Requirements Specification.	(Re-)Read Student's Brief in detail. Company name and logo. Role and responsibility definition. Set-up Project Binder. Start Individual Diary, devise Individual plan of effort in the project.	None
1, 2	(week 2) **29 January - 2 February:** Interviews (some are scheduled at the end of Week 1; all will be completed by the end of week 2) **2 February:** *Teaching Session:* • Briefing on High-Level Design, • Briefing on Fundamental *Agile methods* • Test – Driven Development • *Q&A session*	Interviews with customer - further requirements elicitation. Students **must** attend the respective interview. See the schedule of interviews on Moodle.	None
3–5	(week 3) **9 February:** • Using Visual Paradigm to capture Software architecture (Dental Pro example) • Briefing on Peer-Review • Q&A Session. (week 4) **16 February:** *Teaching Session:* • Java Fragments (GUI, JDBC) • Q&A Session. (week 5) **23 February:** Q&A Session **<u>Friday, 27 February, 5pm</u>** (end of week 5)**:** *Submission* of: "*Requirements Specification and High-Level System Design*" document, and Project Binder and Individual Diaries.	Specifying subsystem requirements, system architecture, and hight level subsystem design. Project Binder and Individual Diaries maintenance.	"*Requirements Specification and High-Level System Design*" document – formal, summative assessment to be submitted on Moodle. Individual Diaries and Team Project Binder submitted via Moodle. Will be reviewed by your ***consultant*** and informal (formative) feedback will be provided.

6–12	(week 6) **2 March:** *Teaching Session* • Briefing on System Implementation • Q&A session (week 7) **9 March:** *Teaching Session* • *Database modelling recap and Q&A Session* • Q&A session. (week 8) **16 March:** *Teaching Session* • Java GUI and Database Connectivity programming (JDBC) recap. • Version Control Tutorial. • Q&A session (week 9) **23 March**: *Teaching Session* • *Feedback on 1st team deliverables (common problems, etc.)* • Test-Driven development, unit and component/subsystem testing with jUnit. • Reverse engineering of Java code with Visual Paradigm. • Q&A Session. (week 10) **30 March:** *Teaching Session* • Preparation for Software Demo • Q&A session. (week 11) **6 April:** Easter Monday (week 12) **13 April:** Q&A session • Software Demo, Implementation and Individual Report **16 April (whole day):** *Demonstration* of the software in operation – formative assessment. **<u>Sun 19 April, 5pm</u>** • *Submission* of Implementation Reports • *Submission* of Individual Reports. • Final versions of Project Binder and Individual Diaries, and any final evidence about individual contribution by any of the team members!	System Implementation and Testing. Attendance to the Demo and Showcase event. Preparation and submission of Implementation Reports. Preparation and Submission of the Individual Reports Project Binder and Individual Diaries maintenance, and submission.	Formal, summative assessments: 1. *Working system demonstration ("Demo")* 2. *Implementation Report* 3. *Individual Report* Individual diaries and Team Project Binder (***to your consultant***)

Table 1: Team Project timetable of briefings, activities and deliverables

Formal Briefing materials (slides) will be prepared and presented in class during the module. The aims of these materials are to introduce what is expected at each stage in the project and to give *some* guidance on how to tackle it. The briefings are ***not a replacement for formal software engineering or OO analysis/design/programming lectures – most, if not all, of the material needed for Team Project will have been covered in other modules***! The timetable for the ***briefing sessions*** and Q&A sessions is as follows:

- Week 1 (26 January) – Introduce Team Project, its structure, learning objectives, assessments, etc.
- Week 1 (26 January) – What is a requirements specification and software architecture?
- Week 2 (2 February) – *Essentials of Agile Software Development? Q&A session, too.*
- Week 3 (9 February) – *Peer-Review and demonstration of using Visual Paradigm to model software architecture. Q&A session, too.*
- Week 4 (16 February) – Java Fragments (GUI, JDBC). *Q&A session, too*
- Week 5 (23 February) – Q&A session.
- Week 6 (2 March) – What is software implementation? (Briefing). *Q&A session, too.*
- Week 7 (9 March) – Java Fragments (2). Version Control. *Q&A session, too.*
- Week 8 (16 March) – Test-Driven Development with jUnit Tutorial, and Q&A. Version control tutorial. *Q&A*

session, too.

- Week 9 (23 March) – General Feedback on first deliverable. Test-Driven development with jUnit. Reverse Engineering with Visual Paradigm. *Q&A session, too.*
- Week 10 (30 March) – Preparation for the Demo *Q&A session, too.*
- Week 11 (6 April) – Easter Monday (no classes)
- Week 12 (13 April) – Q&A session on Demo, Implementation and Individual Reports.

Week 12 (16 April) – Software Demo Day.

The project will proceed according to the following stages:

1. *Team formation and project definition*

   During the first class, the lecturer will present the purpose, organisation and timetable of the team project. The students will be informed of their membership in pre-selected teams and groups. ***This is non-negotiable***!

   Teams will decide upon a company name and logo to appear on all their correspondence and deliverables.

   Each team must set up their Project Binder. Team members' roles and responsibilities will be documented in this binder. The team must decide when these roles are most prominent within the project. Each team member is expected to deliver 100% commitment to the team according to the roles assigned.

2. *Requirements Specification and High-Level Design*

   The customer requirements for the project will be presented. There will be opportunities for questions.

   Teams/Groups must prepare for, and attend the interviews, with the representatives of the fictitious Customer.

   UML (use case diagram/specs) or user stories will be used at this stage to capture the IPOS functional requirements.

   A *requirements specification* will be developed.

   The software architecture of the system will be developed. The decomposition of the IPOS system into sub-systems will be given to Groups, together with a skeleton of a UML component diagram. The teams will need to identify the interfaces between the subsystems and develop detailed specifications for the provided and required interfaces that each of IPOS subsystems will use together with the necessary implementation classes. These will be documented in a simplified design class diagram. Java code will be generated from the defined interfaces and a ***test plan*** for the interfaces/implementation classes will be developed. A high-level system *design document* will be developed.

   One document, containing both the IPOS requirements specification (see above) and IPOS high-level system design, is to be submitted as the ***first summative assessment*** deliverable of the module.

3. *Implementation*

   The system will be coded in its entirety and tested to establish whether it meets the requirements. Each team of a Group will implement the subsystem allocated to the team to develop. Each team will also provide the other teams in the Group the source code of the interfaces and classes necessary for access to the functionality of the subsystem the team has developed (i.e., if Team A of the Group developed sub-system IPOS – SA, then Team A will provide the other teams with the source code of interfaces/classes necessary for other teams to access part of the functionality of sub-system IPOS – SA)[6]. Also, an Implementation Report will be prepared which should include:

   - description of how the system must be compiled (if applicable), deployed, and executed. Using the UML diagrams related to the implementation phase (such as component and deployment diagrams) is recommended but is not mandatory.
   - testing report/plan for unit testing and component/subsystem testing of own subsystem will be developed and completed. A testing plan for the access to the functionality of the other sub-systems will be developed, too.
   - commented source code, and
   - A full design class diagram of the subsystem developed by the team will be submitted as an Appendix to the Implementation report. This diagram may be derived via reverse engineering from the source-code of the subsystem developed by the team.

   The working system will be evaluated via demonstration of the final product ("Demo"), which mainly consists of checking if the functional requirements stated by the Customer are met. Usability of the software will be evaluated too.

   Implementation Reports are included in this phase too.

[6] The teams are free to agree with the other teams how the access to other subsystems will be implemented. A plausible option would be to agree to grant access at the level of database tables, thus making it possible for each subsystem to operate "autonomously" irrespective of whether the other sub-systems are running or not.

4. *Individual report*
   Each student will reflect individually on what has been achieved in the project. He/she will report on lessons learnt and problems experienced, and describe their own, and other members', contribution to the project.

4. Project Assessment

The following is the table of all coursework components to be delivered by each team (and individual), together with the respective proportion of marks:

Item	Description	Week	Mark (%)
1	Requirements Specification and High-Level System Design document	5	15
2	Product Implementation and delivery	11,12	60
3	Individual Report	12	25
	Total		100

Table 2: Team Project Deliverables and Marking scheme

Any team/individual that fail(s) to meet a deadline and have not been granted extension after following the EC/extensions process, as specified by the relevant department's and the university's regulations, will be awarded ZERO marks.

The reasons for the marks awarded will be communicated to the teams via ***Moodle Grader Report*** in a timely manner so as not to unduly penalise their performance in the next stage. For instance, a team that submits a poor high-level design and so receives a low mark will be given feedback about what errors/omissions were made so that they have a chance of producing a correct implementation. There will, however, be **no re-assessment of revised documents** to take account of the feedback provided after the marking has taken place. Producing a document revised in line with the received feedback is ***not required***.
Please note that feedback will be given throughout the module, but especially after each phase, e.g. in the class (general feedback), during Q&A sessions, during lecturer's Office Hours, in private communication with students, etc.

Where members of a team find the contribution of a member to be unsatisfactory, the team is encouraged to resolve the issue internally by negotiation and constructive discussion. If the problem cannot be resolved internally then the team should ask the consultant to intervene. Depending on the circumstances, the consultant might decide to involve the lecturer(s)/module leader(s) in some cases.

In addition, the consultant might during the project, based on clear evidence from a variety of sources (see below), communicate to her/his respective teams and individual team members any concerns known to him/her. The **students are, however, responsible** for informing the consultant of all relevant affairs so that the consultant can form a full and correct view.

VERY IMPORTANT
At the ***end*** of the Team Project the ***final mark allocated to each individual team member for team deliverables will be adjusted in line with the individual's contribution***, despite it being based on the overall marks achieved by the team. In this way, we guarantee fairness to all: non-attendance, breached agreements, lack of contribution etc., can be recognised and the individual final marks awarded to the individuals after adjusting them in line with their respective relative contribution. All adjustments will be justified and backed up by objective evidence such as:

- depositions by team co-workers.
- attendance and involvement in the team's meetings, and the meetings with the team consultant.
- notes from the team meetings with their consultant.
- depositions in the Individual Reports.
- depositions in the Individual Diaries and Project Binder.
- depositions from various forms of communication among team members, e.g. via Moodle, email, social media.
- Etc.

Any amendments to individual marks are at the consultant's discretion (and in agreement with the module leader) and are based on provided evidence by all students by a fixed, announced in advance, deadline.
This procedure guarantees **a fair marking approach for each team member**!

5. Submission of Deliverables

For summative assessment, there are two documents[7] to be delivered by each team ("Requirements Specification and High-Level System Design" deliverable, and "Implementation Report" deliverable), and one to be delivered personally by each student ("Individual Report" deliverable).
All summative assessment documents must be submitted ***electronically via Moodle*** **by the respective deadline**.
The dates for submitting deliverables will appear on Moodle in due time and are tentatively listed in Table 1. They must be strictly adhered to.
Deliverables will be marked, and feedback given ***no later than three, or four, weeks*** of the submission, as specified by the relevant regulations for project, coursework-only modules. Please see https://staffhub.city.ac.uk/policies/student-and-academic-services-policies/assessment-and-feedback-policy and https://www.citystgeorges.ac.uk/about/governance/policies/student-policies-and-regulations.
In addition, after each of the two phases – Requirements & High-Level Design, and Implementation – each team must submit the Project Binder, and each student must submit the respective Individual Diary to the team consultant. Project Binder and Individual Diaries are not marked, but they do serve an important purpose: they are used as evidence when assessing each student's individual contribution (see e.g. Section 4 above).

6. Assessment Criteria for Deliverables

6.1 General criteria for documents

Some written documents submitted for marking will be assessed using the general criteria covering presentation and style. In addition to this, each deliverable will be assessed according to the criteria covering its content as listed below. Fairness of the marking will be ensured by the same section of all deliverables being marked by the same marker(s). For Requirements Specification & High-level System Design document, Implementation Report and Individual Reports, 10% of the overall mark will be awarded for General Criteria.

General Criteria:

- Appropriate title, page numbering and version control (2)
- Introduction, including Purpose & Scope, of the document (2)
- Use of language appropriate to audience (consultant vs. customer), and Spelling and Grammar (2)
- Clear layout and structure (2)
- Appropriate use of graphics and diagrams (2)

Total: 10%

6.2 Requirements Specification and High-Level System Design (Due: Week 5, 15% of total project mark)

This deliverable will be assessed against the following criteria:

- Requirements Specification
  - Full functional requirements specification of the subsystem assigned to the team to develop. (35 marks)
    The teams can use ***one of the following two forms*** to capture the functional requirements:
    - UML Use case model, including:
      - a use case diagram defining the actors, the use cases and their relationships covering the whole functionality of the new system. The other subsystems of IPOS, whose development is delegated to the other teams in the same Group, should appear as actors in the use case diagram.
      (15 marks)
      - Use case specifications for all **use-cases**. The specifications should be concise (as mandated in the UML standard) and provide a ***short description*** of the interactions between the actors and the modelled subsystem, identifying the ***important alternative flows***. You are not expected to use the template for use cases specification used in IN2013. Use instead the template included in the Briefings on Requirements and High-Level Design Document.
      (15 marks)
    - User Stories, the method of documenting functional requirements used in agile methods of software developments. Details on how to write user stories will be provided in the dedicated tutorial session in the module (Week 2).
      (30 marks)

[7] The teams will also need to prepare the summative assessment that is not a document: Demo of the working product.

- Indexed list of all use cases/user stories prioritised according to users' priorities and impact of projected risks during development (i.e., time and budget problems). This should help drive the evolution of the design and implementation tasks.

(5 marks)

- High-Level System Design (50 marks)
  - Develop High-level system design of the entire IPOS as a ***UML component diagram*** in which the three sub-systems (see Section 8 below) are modelled as components (with stereotypes <<subsystems>>) and the interfaces necessary for sufficient communication between the subsystems, as required in the case study.

    (10 marks)
  - The component diagram is supplemented by a fragment of design class diagram, in which the "provided" and "required" interfaces are modelled in detail: i) each method of an interface is fully defined with visibility, scope return type, parameters with types, ii) each method of an implementation class is fully defined with the same level of details as interfaces, but may also include additional methods (not defined in the interfaces) and must include one or more constructors.

    (15 marks)
  - Generate Java code for the interfaces and implementation classes defined in the fragment of design class described above.

    (10 marks)
  - outline (e.g. in a table) how the interfaces between the subsystems can be tested (e.g. using jUnit framework). More specifically, develop test cases for all methods of the interfaces (provided and required) for the subsystem your Team is to develop. The tests should test the correctness of implementations with valid and invalid parameters (e.g., by raising an exception).

    (15 marks)

- Peer-review (5 marks)

  The 1st Team deliverable will be subjected to a peer review by another team in the same Group. The peer review should address the following aspects:
  - Completeness of functional requirements
  - Adequacy of prioritisation
  - Quality and system architecture
  - Quality of the interfaces
  - Adequacy of the interface tests.

  The peer review should provide a concise and ***objective and meaningful*** acknowledgement of the strengths and weaknesses of the 1st deliverable. The peer review should not exceed 2 pages of 10 pt Times with standard page margins. The review should be submitted within the dedicated ("Peer – review Submission Area") ***no later than 7 days*** after the Team has received the 1st Deliverable by another team for review. The team is responsible to organise an efficient peer review (initial draft, internal discussion and comments on drafts). It might be a good idea to designate a team member, e.g. the Deputy Manager, as leading on this task.

  The peer review will be assessed by the Team consultant. The outcome from this assessment (mark and feedback) will be released together with the marks/feedback on the 1st Team deliverable.

Total: 90%. The remaining 10% of the marks for this deliverable will be awarded for the General Criteria.

The target audience for this deliverable is the customer *and* the team consultant. It should provide explanatory material in layman's terms (i.e., use cases/use stories) complemented by technical specifications represented in a structured fashion (e.g. component and class diagram) and source code of the generated interfaces/implementation classes.

6.3 Working Product (Due: Week 11 and 12, 60% of total project marks)

A working implementation of the IPOS system will be demonstrated by each group in a "Demo". Each team of the group will demonstrate the functionality of the sub-system the team has implemented and the access to the functionality of the other two subsystems of IPOS implemented by other teams in the group.
Each team will develop also an "Implementation Report" in which the team documents the implementation related activities, which are detailed below.

A copy of the fully commented source code, or in the case of code generated automatically the appropriate files used by the code generating software, should be submitted in the dedicated submission area on Moodle, as a .zip file. ***Failure to supply the source code will result in Zero marks being awarded for the whole of the Working Product stage.***

The working product will be assessed using the following criteria:

- ***Working product Demo*** (80 marks)
  - Completeness and quality of the product in relation to the requirements (75 marks)
  - Ease of use and consistency of the GUI (5 marks)
- ***Implementation Report*** (20 marks)
  - Software Compilation/Run-time components (3 marks)
    Teams using code generation environments (e.g. NetBeans, Eclipse, IntelliJ IDEA, Visual Studio etc.) should describe the structure of their implementation by referring to the following:
    - Forms and their properties
    - Controls and dialog boxes, menus etc., and their properties
    - Database tables
    - Coded modules
  - Deployment of run-time software components on hardware (2 marks)
  - Testing plans and testing results (e.g. from using jUnit) (12 marks)
  - (Appendix) Design class diagram derived via reverse engineering the source code (3 marks)

During the working product "Demo", a checklist will be used by staff to award marks for the presence of features in the product prototype which are stated in the requirements, and the quality of implementation of those features. Functionality will be assessed using ***two sets of scenarios***, which describe the expected functionality of the product. These scenarios, which will be executed with the completed product after its deployment on a target machine, will be made available to the teams as follows:

i) The first set of scenarios will be given to the teams **at least two weeks before the final demo**. The teams must use this set to prepare for the final demo. Before the start of their final demo the teams are expected to have cleared the database used by the application and to have executed the entire first set of scenarios. ***At the start of the "Demo" assessment the database used by the application should contain the data generated by executing the scenarios included in the first set and nothing else***.

ii) At the "Demo" each team will be presented with another set of scenarios, similar to those included in the first set distributed in advance and asked to execute them with their product.

The marker of the Demo will ask the demonstrator(s)/team to show features in accordance with the requirements.
The demos of the teams in the same Group will be scheduled in the ***same time slot***, and the teams will be taking turns to demonstrate the features of their respective subsystems.
After the features of each subsystem have been demonstrated, the marker will ask for the cross-subsystem interaction/communication to be demonstrated, too. The teams will take turns demonstrating how their respective subsystem interacts with the other two IPOS subsystems.

The software demonstration ("Demo") is planned to take place in week 12 (likely to be on the 16th of April). The details about the Demo, (including the schedule, etc.) will be ***confirmed about two weeks before the demonstration***.

The target audience for the product is the customer, except for the Implementation Report, which is for the consultant.

6.4 Individual Final Report (Due: Week 12, 25% of total project marks)

The target audience for the Individual Report is the consultant. Students ought to use the teams' Project Binder and the respective Individual Diary as inputs when writing the Individual Report. The individual report will be assessed using the following criteria:

- Table of the effort expended in each phase against estimated effort. In case of (substantial) discrepancy an explanation must be provided. Details need to be provided about *own* contribution to the team deliverables. (25)
- Description of how the project went, and a statement of any problems *personally encountered* and how they could be avoided in future. (15)
- Description of how the team worked together, and providing evidence-based assessment about *the contribution of the other members* to the team deliverables. (15)
- Lessons learnt. (20)
- Statement of what you would do differently in future. (15)

Total: 90%. The remaining 10% of the marks for this deliverable will be awarded for General Criteria.

7. Originality of Work

Although the Groups are starting from the same statement of requirements, each Group is expected to work independently. Teams within the same Group are expected to collaborate with the other teams in the Group. Obviously, there will be overall similarities in the deliverables from different teams, and there will be plenty of opportunities to see what other teams are doing, either in the class/Q&A sessions or in private discussions between the members of different teams.

However, copying of (parts of) documents or software of one team by another is, of course, not allowed, and will be treated as plagiarism of coursework!
If copying is detected[8], all students/teams involved will be given **zero** marks for the copied deliverables.

8. InfoPharma Ordering System

Your company has been invited to develop an "IT system to assist with the distribution of pharmaceutical goods" for InfoPharma Ltd, a major pharmaceutical distributor. InfoPharma intend to optimise their interaction with their clients, retail merchants (individuals running own chemist shop and large retailers with chains of stores, e.g. Boots and Superdrags), who sell drugs and other pharmaceutical goods to the members of the public. The new system will also allow small retailers to sell pharmaceutical goods online.
Currently, InfoPharma communicate with merchants by distributing a catalogue of goods available to the interested parties. The catalogue is printed monthly, but in case of major changes of prices special editions can also be printed between the regular updates. Orders are taken by using conventional means such as phone and more recently – e-mail.
Maintaining this way of interacting with the client base is not justified any longer since most of the retailers are already online and have converted parts of their business to 'paperless' technology. Deploying a B2B suite of software applications seems long overdue. InfoPharma want a smooth transition from the current catalogue-based operation to operation relying on a new software order tracking system (InfoPharma Ordering System, IPOS). They would like to have in place for a few months the current operation (catalogue and orders by phone/email) together with IPOS which would allow retailers to have access to the most up to date catalogue and easy ordering from a desktop PC.
In addition, in order to motivate the clients (mainly the pharmacy shop owners) to switch to IPOS quickly, they would like IPOS to include extensive functionality which can be useful to the retailers in their work with their customers: controlling the local stock availability in shops, maintaining customers database and ability to provide discounts to regular customers. IPOS will also allow the shop owner to offer their products online.
InfoPharma Ltd. may decide to offer your company the development of the ***production*** version of IPOS and a maintenance contract for it, if you do a good job with the ***prototype*** of IPOS.

Your consultant has won the contract from InfoPharma's head office to assess their requirements and produce a project plan for the implementation of *one system for one Pharmacy*. Your team has been tasked with doing the job. She/he asks you to set up an appointment for an interview with Mr. Lancaster, the founder and CEO at InfoPharma Ltd. Mr Lancaster has agreed to provide clarification on the requirements for the new system.

If the project plan, the accompanying schedule and the requirements documents are accurate, and the cost is acceptable you will be invited to complete the contract and implement the prototype.

Initial Requirement
The InfoPharma Ordering System (IPOS) will keep track of the orders of pharmaceutical goods made by merchants to InfoPharma Ltd and will assist the merchants' sales to consumers.

IPOS includes ***three*** subsystems:
i) A server application (IPOS-SA), which in the fully functional production version will be administered by the IT department of InfoPharma Ltd.
ii) A client application (IPOS-CA), a copy of which will be run by the merchant on a desktop PC in the merchant's Pharmacy shop.
iii) An online platform (IPOS - PU) to be used by a merchant for direct sales of pharmaceutical goods to the public. The platform will allow the merchants to run advertisement/promotion campaigns with new products with discounts.

In the prototype of IPOS the three subsystems should be implemented as separate ***desktop applications*** able to interact with one another as necessary to meet the requirements. For instance, online sales (via IPOS - PU) should be linked to the inventory of the merchant: successful online purchases should be deducted from the stock of goods available at merchant's shop. Likewise, orders from IPOS-SA and delivered to a merchant, should be easy to record in IPOS-CA and increase the merchant's stock. Finally, promotion campaigns run via IPOS – PU may affect the cost of selling goods online (at reduced price), but not the cost of the same goods if sold in the pharmacy shops.

8.1 IPOS-SA

This subsystem should allow InfoPharma Ltd. to maintain an electronic catalogue of pharmaceutical products and control stock availability. The details of an item in the catalogue (products ID, product description, unit price and

[8] Checks will be made! For example, via automatic comparison of source code submissions, etc.

current availability) are detailed in Appendix 1. The catalogue is accessed by merchants who should have valid user accounts set for them by the administrator of IPOS-SA. A valid user can access the catalogue and place an order by filling in an order form (Appendix 2). Once an order is submitted IPOS-SA should generate an invoice which will be stored in the system database for further reference and can be displayed on screen or printed out by either the corresponding merchant or by the Operation's Manager at InfoPharma Ltd. The status of the orders (accepted, being processed, dispatched) will be maintained by IPOS – SA and members of staff at InfoPharma should be able to update it when needed (e.g., when an order is dispatched a member of the dispatching department will have to enter the following details: who dispatched it, when was the order dispatched, the courier used for the delivery, courier's Ref. No, expected delivery time). The merchant should be able to track their orders through IPOS, too.
The merchants are given credit by InfoPharma Ltd. and can make payments for the goods they have ordered by the *end of the calendar month*. They can check their balance by accessing IPOS_SA. Payments by merchants are not part of IPOS-SA and should be made using other means, such as direct bank transfers using InfoPharma IBAN number shared with all registered merchants. Once a payment is received, payment details should be entered into the system by the accounting department at InfoPharma, and the account balance of the respective merchant - will be updated[9]. If a merchant is late no more than 15 days with their payment, they can continue to use the ordering system, but IPOS-SA will generate a reminder on the screen every time the account is accessed by the merchant. If a merchant is late with a payment between 15 days and 30 days, then IPOS-SA will suspend the user account, i.e. no further orders will be accepted. Once a payment is received within 30 days of the payment deadline the 'suspended' account is restored to 'normal' by the system and the user can resume placing new orders. If the payment delay is longer than 30 days, the user merchant account is flagged as 'in default'. Restoring an account from an 'in default' to 'normal' (or 'suspended') state can only be done after an authorisation by the Director of operations of InfoPharma Ltd.

The functionality of IPOS-SA can be divided into the following packages:

IPOS-SA-ACC: This package provides the functionality needed for setting up accounts for the *users* of IPOS-SA. Different users may have different privileges and use different parts of IPOS-SA. At least the following three groups of accounts must be implemented in the prototype:

i) a **merchant's account**. The user with such an account should be able to access the following packages: IPOS_SA-CAT and IPOS-SA-ORD (see below). Once a new account is set up as a merchant account the system will ask for contact details to be provided, credit limit and discount plan to be set up for the new merchant before the account is activated. If the required details are not provided the account will not be created. The discount plan sets discount rates which will be applied to orders for respective merchant. At least the following two types of discount plans must be implemented:

- *fixed discount plan*, i.e. the same discount rate applies to all items of any quantity ordered from the catalogue. The discount is applied to each order and the amount due is reduced by the amount of the discount at the time of placing an order.
- *flexible discount plan*, in which the discount rates depend on the value of all orders *within a calendar month*, e.g. 1% for orders less than £1000, 2% for orders between £1000 and £2000 and 3% for orders in excess of £2000. With the flexible discount the value of the discount is calculated at the end of the calendar month. The amount of the discount is either paid back to the merchant by sending a cheque at the end of the month or deducted from the value of the next order(s) by the same merchant.

ii) an **administrator's account**. This account has full access to the functionality of IPOS-SA, i.e. access to all packages of IPOS-SA including IPOS-SA-ACC.

iii) a **manager's account**. This account allows the user an access to IPOS-SA-RPRT. This user can also access the merchant accounts and alter their credit limits, discount plans and change the state of an 'in default' account to either 'normal' or 'suspended'.

IPOS-SA-CAT: This package contains the functionality necessary to maintain a catalogue of goods available to the merchants to order from. As a minimum the following functions are required:

- creating a (new) catalogue (optional, if it does not already exist. It is acceptable not to have this function as part of IPOS-SA, in which case the catalogue should be created using tools available for administering the IPOS-SA database),
- adding a new item, deleting an item, updating an item, searching for an item using various searching criteria (item ID, key term, etc.),
- adding new stock of a catalogue item (e.g. when they are delivered to InfoPharma). For each item in the catalogue a minimum stock level parameter can be set up (a non-negative integer). The system must warn the relevant users (administrator and the operation's manager) every time they use IPOS-SA about the occurrence of low stock conditions. A report should be

[9] In the full production version of IPOS payment may be integrated with on-line banking, i.e. the merchants may be able to make payment by bank transfers. In the prototype, however, the payment details are expected to be entered into the system manually.

possible to generate on demand to list all items in the catalogue, which are below their set minimum stock levels (see Appendix 3).

IPOS-SA-ORD: This package contains the functionality necessary for a merchant to be able to place a new order (choosing items from the current catalogue) and track the status of their orders, the outstanding unpaid balances, etc. Once a new order is accepted by the system, the InfoPharma stock should be reduced accordingly. Raising invoices against accepted orders and entering data about the payments made for previous orders (bank transfers, card or cheque payments) is also part of this package.

IPOS-SA-RPT: Report generating package. This package should allow various reports to be generated by searching the system database. The report should be visualised on computer screen and printed (when required). The prototype should allow for generating the following reports:

i) Turnover for a *given period of time* in terms of quantities of goods sold to merchants and the revenue received by InfoPharma Ltd.

ii) List of orders received from a particular merchant for a given period of time with the following details (see Appendix 4):
- order ID,
- date of ordering,
- value of the order,
- date of dispatching,
- payment received (or pending),
- a 'Totals' line should conclude the report.

iii) Report about the activity of an individual merchant for a given period of time with the following details (see Appendix 5):
- contact details of a merchant (as a header),
- list of all orders (including the individual items ordered, quantity ordered, individual cost, total cost of order, discount given if known, payment status).

iv) List of invoices raised against an individual merchant for a given period (on the screen) and then navigating through the details of the individual invoices. Printouts of all these should also be possible to generate.

v) List of all invoices raised by InfoPharma Ltd against merchants for a given period.

vi) Stock turnover (goods sold and newly received) within a given period of time.

8.2 IPOS-CA

This subsystem is a desktop application, which implements a combination of features allowing the merchant to access IPOS-SA-ORD, and a set of additional features to automate the operation of the merchant in dealing with their own customers. More specifically the functionality of IPOS-CA includes the following packages:

IPOS-CA – ORD: This package allows the merchant to place an order with IPOS-SA, track its progress through IPOS-SA (shipping, delivery), view the previous orders and query own outstanding balance. A merchant is expected to access *IPOS-SA-ORD* by using a username and password created for them by the administrator of *IPOS-SA* and sent to them using a reliable delivery means, e.g. a recorded post.

IPOS-CA-USER: This package should implement a light touch authentication (a username/password) with role-based access to the functionality of IPOS-CA. As a minimum the following roles of users must be implemented: i) Pharmacist, the user who can use all packages except IPOS-CA-USER, ii) Admin will be able to deal with user accounts for IPOS-CA (create a new user account with a specific role, and remove an account); iii) Manager – this role should be able to generate the reports envisaged in IPOS-CA-RPT, below.

IPOS-CA-CUST: Maintain a database of account holders (consumers regularly buying pharmaceutical and beauty products from the merchant)[10]. Account holders are given credit up to a predefined limit, can be given a discount (the same two types of discount plans as for IPOS-SA must be implemented – fixed and flexible) on the purchases, and are allowed to pay by the end of the calendar month. Between the 5th and 15th of every month IPOS-CA should allow the merchant to generate statements for all those account holders who have not cleared their balances at the end of the previous calendar month. In the statement the account holders will be asked to pay in full by the 15th of the current month. This feature will be used on demand. If account holders are late with their payments, the system should allow the merchant to produce two reminders (see Appendix 6) about late payment, respectively, as follows.

[10] In the production version the account holders will be given a physical membership card, which they can use to confirm their identity when buying goods in the pharmacy. In the prototype, the account holders will confirm their membership by providing the pharmacists with the membership account number.

- IPOS-CA should maintain ***automatically*** (i.e. without human intervention) the status of an account according to Fig. 1 below. The diagram shows how the status of an account changes from 'normal' (assumed in February), to 'suspended' (in mid-March) and finally to 'in default' (end of March).

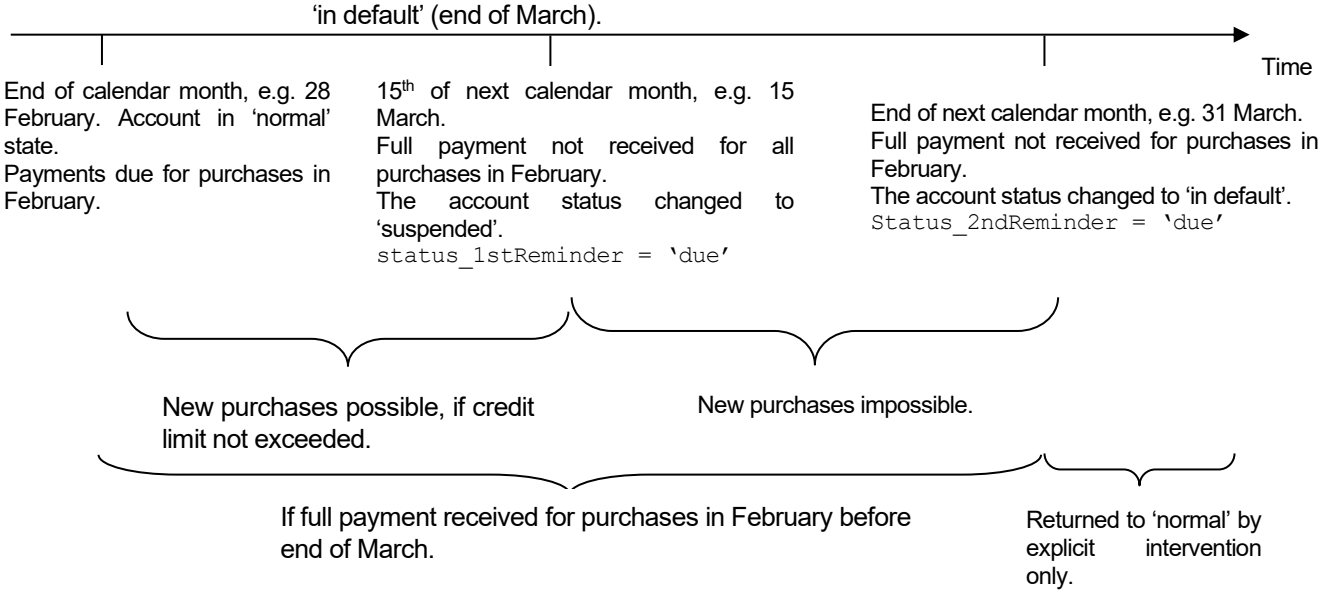

Fig. 1. Illustration of how IPTS-CA changes the status of an account.

- The reminders to debtors are generated ***on demand*** (i.e. explicitly by the user of IPOS-CA). This functionality can be used at any time. There are two types of reminders, 1st reminder and 2nd reminder, with slightly different wording of the text as suggested in Appendix 6. Each of these reminders is generated only *once* when a set of conditions are met, as defined below. There can be many 1st and 2nd reminders sent to the same account holder, but they should refer to *different debts*. Between two 1st reminders sent to the same account holder, for example, the account will have its status changed from 'suspended' or 'in default' to 'normal' and then back to 'suspended'/'in default' as shown in Fig. 1. Synchronisation between the two asynchronous processes – changing the account status by IPOS-CA and generating reminders by a human user – can be achieved by using two attributes associated with an account:
  - `status_1stReminder` and `status_2ndReminder`, of an account with respect to the 1st and 2nd reminders, respectively, will take their values from the set `{no_need, due, sent}`. `'no_need'` indicated that no reminder of the particular type needs to be generated when the function of generating reminders is used, `'due'` indicates that a reminder should be generated, `'sent'` indicates that the reminder of the particular type (1st or 2nd) has been sent (e.g. by post as "recorded delivery" letter) and there is no need to send it again before the account status changes. When an account is created, its status is initialised to '`no_need`'.
  - `date_1stReminder` and `date_2ndReminder`, associated with each of the accounts, specify the time when a reminder should be sent. The values that this attribute can take are either `'now'`, i.e. immediately, or a particular date, e.g. `21-Mar-2025`, which indicates that the reminder should not be generated *before* this date.

    The pseudo-code of an algorithm generating the reminders is shown below:

```
if (status_1stReminder = 'due')
  {
     generate(1st Reminder);  // Payment date must be set to <current
date> + 7 days
     status_1stReminder = 'sent';
     date_2ndReminder = <current date> + 15 days;
     // Second reminder scheduled to be sent eventually 15 days later
  }
if (status_2ndReminder = 'due')
  {
     if (date_2ndReminder <= <current date>)
        {
           generate(2nd Reminder);    // Payment date set to <current
date> + 7 days
```

```
            status_2ndReminder = 'sent';
        }
  }
```

When a *full payment* is received (which can take place at any time), the following algorithm should be applied:

```
if (account_status != 'in default')
  {
     status_1stReminder = 'no_need';
     status_2ndReminder = 'no_need';
  }
```

IPOS-CA-Templates: This package allows Managers to change the templates used for reminders and receipts/invoices and details related to the merchant identity (e.g. Pharmacy name, logo, address and email)

IPOS-CA-Stock: This package should also maintain a database with the local stock availability and provide warning of low stock to help the merchant order in due time new stock from InfoPharma. The structure of the stock records should be similar to the catalogue structure in IPOS-SA except that an extra field will be added to carry the retail mark-up rate to be applied to the bulk cost of the item (as received from InfoPharma). VAT will be also added at a rate common for all goods. This parameter can be set to 0% but may change and therefore must be configurable[11].

IPOS-CA-Sales: This package allows for sales to customers (account holders and occasional customers) to be recorded in IPOS-CA including payments. A consumer can make payments using *cash* and *credit/debit cards*. The system should store the form of payment and the amount received. In case of card payments, the card type, the first four and the last four digits of the card number and the expiry date should also be stored. Account holders are only allowed to use cards (credit or debit) for their payments. Occasional customers (i.e. who do not have an account with the merchant) can use both forms of payment, but their payments should be made in full at the time of purchasing pharmaceutical goods. A successful sale leads to producing a receipt/invoice (see Appendix 7).

IPOS-CA-RPT: This package allows Manager to generate the following reports:

- turnover within a given report period (amount of sales, value of new orders placed with InfoPharma),
- stock availability report: items, availability, value of stock (and VAT).
- change of aggregated debt by account holders for a given period: aggregated debt at the beginning of period (i.e. how much the account holders owed at the beginning of period), payments received from all debtors within period, new aggregated debt from all account holders at end of the period.

8.3 IPOS – PU

This subsystem offers an online portal which a merchant may use to sell goods to members of the public online.

This subsystem should offer the following functionality:

IPOS – PU – Members: The package will allow members of the public to submit a membership application with the online portal. The applicant may select to register as either a "commercial" or a "non-commercial" customer. An application for a commercial membership should include a company registration number (with Companies House in the UK) and additional details on the Company Director(s), type of business, address, email address, etc[12]. Applications for commercial membership will be passed to IPOS-SA[13]. A decision will be taken by InfoPharma Ltd staff after conducting the necessary diligence checks. If the checks are successfully passed, the application for commercial membership will be approved. InfoPharma Ltd will notify the approved applicant by email, in which details on how to access IPOS – SA will be included. The customer may opt to receive the confirmation by regular mail (e.g. via a registered delivery).

Applications for non-commercial membership do not require any personal details to be provided. A decision on the application is taken by the portal after checking the validity of the email address provided by the applicant in the application. In case of successful registration, non-commercial

[11] The current position in the UK on drugs/medications is more nuanced. In the UK, most prescription drugs dispensed in the community are **zero-rated (0%) for VAT**, meaning no tax is charged, especially for NHS prescriptions or private ones for personal use. However, drugs administered by a health professional (like injections) or consumed in hospitals are often **standard-rated (20% VAT)**, though a temporary zero-rate for some patient group directions (PGDs) runs until March 2027, aligning them with standard prescriptions. These details should be ignored in IPOS-CA and a single VAT rate should apply to all drugs sold by merchants.

[12] For full details to be provided with an application for "commercial", refer to the info recorded in IPOS-SA on merchants.

[13] This may seem odd, since the portal is meant to serve a single merchant, typically a small company. The intention with the prototype is to test this functionality without overcomplicating IPOS-SA. The production version may well have this functionality moved to IPOS-SA.

members will be notified by an email in which their login details will be included: the email provided by the applicant will be used as username, while a password, necessary for the login, will be generated by IPOS-PU (a random string of 10 symbols which include letters, numbers and special symbols). The newly registered non-commercial member will be forced to change their password when they login to IPOS_PU for the first time. Non-commercial members will be offered an incentive of 10% discount on every 10th order they make via the IPOS-PU portal.

IPOS-PU-SALES: The package will allow customers to view the catalogue of goods offered by the respective merchant. Filtering of the catalogue should be provided (e.g., to constrain the list to goods to those that match a keyword, e.g. "paracetamol") to narrow down the list of products of interest to the customer. The customer will be able to select from the catalogue the products they require and place them in a shopping cart, confirm the order and checkout paying for them using a credit/debit card. A successful purchase will lead to an email being sent to the customer with details about how to track the order (from being received, to being dispatched and then delivered). The platform should maintain the status of the order until it is eventually delivered. Successful orders via IPOS-PU should be "propagated" to the respective merchant (i.e. IPOS-CA): the merchant's inventory will be affected (deducting the amount of ordered goods).

IPOS-PU-COMMS: This package will offer communication with 3rd party services such as i) an email provider, e.g. Google SMTP, and ii) a Payment Processor, such as PayPal. The integration with the email service will be used by IPOS-PU and will be also made available to the other IPOS subsystems – IPOS-SA and IPOS-CA, to use as necessary. For instance, IPOS-SA will be emailing successful applicants the details of how they can use the respective IPOS-SA services[14].

IPOS-PU-PRM: This package will allow the respective merchant to run "promotion/advertising campaigns" via the IPOS-PU portal. The campaigns will be setup and managed by the Admin of the IPOS-PU subsystem. Each promotion campaign will be set to run from "Start Date/Time" to "End Day/Time" and will include a list of pharmaceutical goods from the Merchant's stock with a fixed discount attached to each item. The same discount can apply to all items included in the promotion campaign, or different discounts can be applied to different items on the list. More than one promotion campaign can be "run" (i.e. be active) simultaneously, as necessary. A promotion campaign runs until the end of the specified period unless it is cancelled by the IPOS-PU Admin. Every time a member of the public visits IPOS-PU, they will be offered a link "Promotions", provided the visit falls within one of the active promotion campaign periods. Clicking on this link will list the active promotion campaigns. The customer can then select one of the active promotion campaigns and look at the products included in it and the discounts offered for different products. The customer can also proceed and compile a new order using the items on the list of products included in the campaign. The package will keep track of the "clicks" of links to each advertising campaign and the items included in the campaign. More specifically, if an advertising campaign CAMP_X contains a list of two items (Item_X_1 and Item_X_2), then IPOS-PU should maintain the following 5 counters: CAMP_X_counter of how many times the link to CAMP_X is clicked by customers visiting IPOC-PU, CAMP_X_1 and CAMP_X_2 counters which will be increased every time a customer includes Item_X_1 and Item_X_2, respectively, in an order, and finally, CAMP_X_1_purchased and CAMP_X_2_purchased, which count the number of items Item_X_1 and Item_X_2 included in completed orders during CAMP_X campaign. For instance, if an order includes 3 items of Item_X_1, then the counter CAMP_X_1 is increased by 3. Likewise, if an order includes 5 items of Item_X_2, then the counter CAMP_X_1 is increased by 5. Should the order be paid for, then CAMP_X_1_purchased and CAMP_X_2_purchased will be increased, too, by 3 and 5 respectively, but will not be increased if the order remains unpaid. Thus, CAMP_X_1 and CAMP_X_2 may be significantly greater than CAMP_X_1_purchased and CAMP_X_2_purchased, respectively. The ratios CAMP_X_1_purchased / CAMP_X_1 and CAMP_X_2_purchased / CAMP_X_1, respectively can be seen as a rough estimate of the "conversion rate" for each of the items included in a promotion campaign. Note that these counters may be significantly greater than the quantity of items which are actually purchased via IPOS-PU since only a fraction of the orders including items at promotional prices will be completed. Tracking the user behaviour (conversion rate), and how successful a particular promotion campaign has been, is achieved via a number of reports listed in the next package.

IPOS-PU-RPT: This package shall allow Admin to generate the following reports:

- Sales within a specified period (quantity of goods sold, unit price per unit and price for the quantity of sales per item). An example is provided in Appendix 8.

[14] Implementation of the integration of IPOS-PU prototype with third party services is **optional**, and if implemented will get the respective team(s) an extra 5% on top of the marks for the implementation of the remaining functionality. In case a full integration with an SMTP server is not implemented, the prototype should record somewhere (e.g. in a databases table) the email details (a timestamp of the record, the recipient email address, the body of the email, and optionally any attachment that may be needed with the emails). In case the integration with PayPal or alternative payment processor is ***not*** implemented, the prototype should store the payment details somewhere (e.g. in another database table) with the following details: payee details, amount, timestamp of the payment, card number used to make the payment.

- Advertising campaigns for a given period of time (see Appendix 9).
- Count of hits of advertising campaigns broken down by goods included in the advertising campaign (see Appendix 10).

9. Documents provided by customer

9.1 Catalogue Layout

Item ID	Description	Package Type	Unit	Units in a pack	Package Cost, £	Availability, packs	Stock limit, packs
100 00001	Paracetamol	box	Caps	20	0.10	10,345	300
100 00002	Aspirin	box	Caps	20	0.50	12, 453	500
100 00003	Analgin	box	Caps	10	1.20	4,235	200
100 00004	Celebrex, caps 100 mg	box	Caps	10	10.00	3,420	200
100 00005	Celebrex, caps 200 mg	box	caps	10	18.50	1,450	150
100 00006	Retin-A Tretin, 30 g	box	caps	20	25.00	2,013	200
100 00007	Lipitor TB, 20 mg	box	caps	30	15.50	1,562	200
100 00008	Claritin CR, 60g	box	caps	20	19.50	2,540	200
200 00004	Iodine tincture	bottle	ml	100	0.30	2,2134	200
200 00005	Rhynol	bottle	ml	200	2.50	1,908	300
300 00001	Ospen	box	caps	20	10.50	809	200
300 00002	Amopen	box	caps	30	15.00	1340	300
300 00003							
400 00001	Vitamin C	box	caps	30	1.20	3,258	300
400 00002	Vitamin B12	box	caps	30	1.30	2,673	300

Note: This is the proposed layout of the electronic catalogue. The left 6 columns are identical to the layout of the current paper-based catalogue. The 7th column, 'Availability' will be shown to the merchants at the time of their placing an order. The right most column, 'Stock Limit' is not meant to be shown to the merchants at the time of placing an order. It is only here to assist InfoPharma with ordering of new stock from their suppliers.

9.2 Appendix 2: Order Form

Client: Cosymed Ltd.
3, High Level Drive
Sydenham, SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

IPOS Account: 0000235
Date: 12/01/2025

Item ID	Description	Quantity	Unit Cost, £	Total, £
100 00001	Paracetamol	10	0.10	1.00
100 00003	Analgin	20	1.20	24.00
200 00004	Iodine tincture	20	0.30	3.60
200 00005	Rhynol	10	2.50	25.00
300 00001	Ospen	10	10.50	105.00
300 00002	Amopen	20	15.00	300.00
400 00001	Vitamin C	20	1.20	24.00
400 00002	Vitamin B12	20	1.30	26.00
		Grand Total:		**508.60**

For Cosymed Ltd:

/Signature/

Note: This is the layout of the order form currently used by the merchants to place an order with InfoPharma Ltd.
A similar layout is expected to be used in IPOS.

9.3 Appendix 3: Low Stock Level Report

Item ID	Description	Availability, packs	Stock limit, packs	Recommended Min Order
100 00001	Paracetamol	250	300	80
100 00002	Aspirin	320	500	230
100 00003	Analgin	102	200	118
200 00004	Iodine tincture	87	200	133
200 00005	Rhynol	201	300	129

Generated: 31/01/2025
By: Director of Operations

Note: This is the proposed layout of the report. The right most column, 'Recommended Min Order' is calculated so that the availability should become 10% above the stock limit. 10% can be set as a parameter for the individual items in the catalogue (e.g. between 10% and 50%).

9.4 Appendix 4: Merchant's Orders Summary

Report Period: 01/01/2003 – 31/01/2003

Client: Cosymed Ltd.
3, High Level Drive
Sydenham, SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

IPOS Account: 0000235

Order ID	Ordered	Amount, £	Dispatched	Delivered	Paid
IP2034	12/01/2003	302.50	14/01/2003	15/01/2003	20/01/2003
IP 2780	17/01/2003	525.00	18/01/2003	19/01/2003	Pending
IP 3021	29/01/2003	750.30	Pending	Pending	Pending
Total:	**3**	**1577.8**	**2**	**2**	**1**

Generated: 31/01/2003
By: Director of Operations

Note: This is the proposed layout of the summary.

9.5 Appendix 5[15]: Merchant's Orders Detailed Report

Report Period: 01/01/2003 – 31/01/2003

Client: Cosymed Ltd.
3, High Level Drive
Sydenham, SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

IPOS Account: 0000235

Order ID	Cost, £	Ordered	ItemID	Quantity	Unit cost, £	Amount, £
IP2034	302.50	12/01/2003				
			100 00001	10	0.05	0.50
			100 00045	20	5.10	102.00
			100 00345	10	2.60	26.00
			200 000056	10	17.40	174.00
IP 2780	525.00	17/01/2003				
			100 00234	10	0.30	3.00
			100 00567	20	10.20	204.00
			300 00089	30	10.60	318.00
IP 3021	750.30	29/01/2003				
			200 00045	30	15.00	450.00
			300 00187	25	10.00	250.00
			400 00234	2	25.15	50.30
Total:	**1577.80**	**3**				

Generated: 31/01/2003
By: Director of Operations

Note: This is an example of the proposed layout of the detailed merchant's report.

[15] The layout is provided without "discount" column, which is implied in the description of the report on Merchant activities (see IPOS-SA-RPT, iii).

9.6 Appendix 6: Payment overdue Reminders

Client: Cosymed Ltd.
3, High Level Drive,
Sydenham,
SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

InfoPharma Ltd.,
19 High St.,
Ashford,
Kent

5th January 2003

Dear Client,

REMINDER - INVOICE NO.: 197362

IPOS Account: 0000235 **Total Amount: 294.00**

According to our records, it appears that we have not yet received payment of the above invoice, which was raised against Cosymed Ltd. on 18th December 2002, for ordering pharmaceutical goods from InfoPharma Ltd, order IP10023.

We would appreciate payment at your earliest convenience.

If you have already sent a payment to us recently, please accept our apologies.

Yours sincerely,

A. Petite, Director of Operations, InfoPharma Ltd.

Client: Cosymed Ltd.
3, High Level Drive,
Sydenham,
SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

InfoPharma Ltd.,
19 High St.,
Ashford,
Kent

20th January 2003

Dear Client,

SECOND REMINDER - INVOICE NO.: 197362

IPOS Account: 0000235 **Total Amount: 294.00**

It appears that we still have not yet received payment of the above invoice, which was raised against Cosymed Ltd on 18th December 2002, for ordering pharmaceutical goods from InfoPharma Ltd, order IP10023, despite the reminder sent to you on 5 January 2003.

We would appreciate it if you would settle this invoice in full by return.

If you have already sent a payment to us recently, please accept our apologies.

Yours sincerely,

A. Petite, Director of Operations, InfoPharma Ltd.

Note: These are the proposed layouts of the two payment reminders.

9.7 Appendix 7: Retail Invoice

J. Smith,
27 Sainsbury Close,
Stratford,
Essex EJ6 5TJ

Cosymed Ltd.,
3, High Level Drive,
Sydenham,
SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

18th December 2002

Dear Mr. Smith,

INVOICE NO.: 197362

Account No: CSM000123

Item ID	Packages	Package Cost, £	Amount, £
Paracetamol	1	0.50	0.50
Analgin	2	3.20	6.40
Iodine tincture	2	0.80	1.60
Rhynol	5	6.50	32.50
Amopen	2	30.00	60.00
Vitamin B12	2	2.90	5.80
		Total	106.80
		VAT @ 17.5%	18.69
		Amount Due	**125.49**

Thank you for your valued custom. We look forward to receiving your payment in due course.

Yours sincerely,

J. Faith

9.8 Appendix 8: IPOS-PU Sales Report

IPOS-PU Sales Report

Start Period: 01/05/2025
End Period: 31/05/2025

Cosymed Ltd.,
27 Sainsbury Close,
3, High Level Drive,
Sydenham,
SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

Item ID	Description	Sold, packs	Unit price, £	Total, £
100 00001	Paracetamol	250	0.10	25.00
100 00002	Aspirin	320	0.50	160.00
100 00003	Analgin	102	1.20	122.40
200 00004	Iodine tincture	87	0.30	26.10
200 00005	Rhynol	201	2.50	502.50
Total online sales for period		960	N/A	836.00

Generated: 30/06/2025
Generated by IPOS-PU Operations

9.9 Appendix 9: Report on Advertising Campaigns

IPOS-PU Advert Campaigns Report

Cosymed Ltd.,
27 Sainsbury Close,
3, High Level Drive,
Sydenham,
SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

Start Period: 01/03/2025
End Period: 31/05/2025
Active campaigns: 2 (Camp 5 and Camp 6)

Campaign ID	Start Date/Time	End Date/Time	Items included	Discount, %
Camp 05	01/03/2025	31/03/2025	3	Fixed, 5%
Sold				
ID	Description	Discount	Items sold	Total Sales, £
100 00002	Aspirin	5%	20	9.50
100 00003	Analgin	5%	100	114
200 00004	Iodine tincture	5%	20	5.70
Total Sales in campaign:				129.20
Camp 6	01/04/2025	15/05/20	2	Variable
Sold				
ID	Description	Discount	Items sold	Total Sales
200 00005	Rhynol	10%	50	112.50
300 00001	Ospen	20%	150	1260
Total Sales in campaign:				1372.50

Generated: 30/06/2025
Generated by IPOS-PU Operations

9.10 Appendix 10: Report of Customer engagement with Advertising Campaigns

Customer Campaign Engagement Report

Cosymed Ltd.,
27 Sainsbury Close,
3, High Level Drive,
Sydenham,
SE26 3ET
Phone: 0208 778 0124
Fax: 0208 778 0125

Campaign ID: Camp 05
Campaign Description: March Advertising Campaign, 3 items, Fixed Discount of 5%
Start Period: 01/03/2025
End Period: 31/03/2025

Counter ID	Counter description	Hits count	Purchases	Conversion Rate
Camp 05	Campaign hits	3000	N/A	N/A
Item(1) hits	"Aspirin" hits	2000	50	50 / 2000 = 0.25 (2.5%)
Item(2) hits	"Analgin" hits	1000	10	10/1000 = 0.01 (1%)
Item(3) hits	"Iodine tincture" hits	200	20	20/200 = 0.1 (10%)

Generated: 30/06/2025
Generated by IPOS-PU Operations

10. Teams, Groups and consultants

The Teams/Group membership is provided in the table below. The mapping between Teams and subsystems is as follows:

- Team A will be responsible for the development of IPOS – SA,
- Team B will be responsible for the development of IPOS – CA, and
- Team C will be responsible for the development of IPOS – PU.

First Name	Surname	Email	Team	Group	Consultant
DAVID	CHERRY	David.Cherry@city.ac.uk	1 Team A	1	Ali Sarvghad Batn Moghaddam Ali.Sarvghad-Batn-Moghaddam@citystgeorges.ac.uk
SHEERAZ	MOHAMMED	Sheeraz.Mohammed@city.ac.uk			
HELIYA	VATANI	Heliya.Vatani@city.ac.uk			
ALI	SALAM	Ali.Salam@city.ac.uk			
TAAHA	AL-BARI	Taaha.Al-Bari@city.ac.uk			
AHMED	ALI	Ahmed.Ali.7@city.ac.uk			
ABDULAZIZ	ALKUWARI	Abdulaziz.Alkuwari@city.ac.uk			
ESHAN	AHMED	Eshan.Ahmed@city.ac.uk	2 Team B		
ASMA	MAHBOOB	Asma.Mahboob@city.ac.uk			
FAWAZ	ALI	Fawaz.Ali@city.ac.uk			
CHANG QI	LIN	Chang-Qi.Lin@city.ac.uk			
SAMUEL	DARKO	Samuel.Darko@city.ac.uk			
BAYAN NASSER S	AL JUBRAN	Bayan.Al-Jubran@city.ac.uk			
ISHMAEL	RICHARDS GLOCKNER	Ishmael.Richards-Glockner@city.ac.uk			
SEYER	RAJI	Seyer.Raji@city.ac.uk	3 Team C		
SOFYA	SHORUNOVA	Sofya.Shorunova@city.ac.uk			
JOYCE	OW	Joyce.Ow@city.ac.uk			
BASIT	HUSSAIN	Basit.Hussain@city.ac.uk			
ETHAN	KAN	Ethan.Kan@city.ac.uk			
DA GAMA	PRESLEY	presley.da-gama@city.ac.uk			
MANASAR	ABDULLAHI	Manasar.Abdullahi@city.ac.uk			
HERMAN	SILDNES	Herman.Sildnes@city.ac.uk	4 Team A	2	Peter Popov p.t.popov@city.ac.uk
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KANZAH	HUSSAIN	Kanzah.Hussain@city.ac.uk			
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YAMIN	AHMED	Yamin.Ahmed.2@city.ac.uk			
JENIS	KHORSHED	Jenis.Khorshed@city.ac.uk			
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LAMIN	CEESAY	Lamin.Ceesay.2@city.ac.uk			

NIKITA	BRYNDAK	Nikita.Bryndak@city.ac.uk	6 Team C		
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AHMED MUSTAPHA	DRAME	Mustapha.Drame@city.ac.uk			
KHALID	ALI	Khalid.Ali@city.ac.uk			
ZAYD	MOTIWALA	Zayd.Motiwala@city.ac.uk			
ADIYAT	NIAZ	Adiyat.Niaz@city.ac.uk			
NAFIJA	MIAH	Nafija.Miah@city.ac.uk	7 Team A	3	~~Walter, Martin~~ ~~Martin.Walter@citystgeorges.ac.uk~~ Peeroo, Kaleem Kaleem.Peeroo@citystgeorges.ac.uk
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SULAYMAN	SALEEM	Sulayman.Saleem@city.ac.uk			
FATTEH	SIKDER	Fatteh.Sikder@city.ac.uk			
NASIF	ALOM	Nasif.Alom@city.ac.uk			
PRION	JALAL	Prion.Jalal@city.ac.uk			
MAHMOUD	ABEDO	Mahmoud.Abedo@city.ac.uk			
ABDULRAHMAN AHMAD A A	AL-MAWLAWI	Abdulrahman.Al-Mawlawi@city.ac.uk	8 Team B		
MARGARITA	TOTEVA	Margarita.Toteva@city.ac.uk			
ALVINA	TARIQ	Alvina.Tariq@city.ac.uk			
DOMINYKAS	MILKINTAS	Dominykas.Milkintas@city.ac.uk			
OMAR	UDDIN	Omar.Uddin@city.ac.uk			
KHALID	ELMI	Khalid.Elmi@city.ac.uk			
ABDULAZIZ A A A	MARAFI	Abdulaziz.Marafi@city.ac.uk			
ALI	LADAK	Ali.Ladak@city.ac.uk	9 Team C		
AMIR	AZOUK	Amir.Azouk@city.ac.uk			
TASAL	ISLAM	Tasal.Islam.2@city.ac.uk			
MUHAMMAD	HAMZA	Muhammad.Hamza.3@city.ac.uk			
YUSUF	AHMED	Yusuf.Ahmed.5@city.ac.uk			
ABDULLAHI	BODLEH	Abdullahi.Bodleh@city.ac.uk			
AFFAN	NAZIR	Affan.Nazir@city.ac.uk			
NEIL	KAPADIA	Neil.Kapadia@city.ac.uk	10 Team A	4	~~Walter, Martin~~ ~~Martin.Walter@citystgeorges.ac.uk~~ Peeroo, Kaleem Kaleem.Peeroo@citystgeorges.ac.uk
YUSRA	ABUBAKAR	Yusra.Abubakar@city.ac.uk			
AADAM	MARWAT	Aadam.Marwat@city.ac.uk			
ARSLAN	SHAIKH	Arslan.Shaikh@city.ac.uk			
ARIUNSANAA	OTGONBAYAR	Ariunsanaa.Otgonbayar@city.ac.uk			
RAYYAN	KHAN	Rayyan.Khan.2@city.ac.uk			
CRAIG	ATAIDE	Craig.Ataide@city.ac.uk			
KAYA	MEHMET	Kaya.Mehmet@city.ac.uk	11 Team B		
NAFEEZ	SYED	Nafeez.Syed@city.ac.uk			
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